

EXERCISE

CONSIDER THE FOLLOWING SYSTEM:

$$x(t+1) = Fx(t) + Gu(t)$$

$$y(t) = Hx(t)$$

WHERE:

$$F = \begin{bmatrix} 1 & 9/2 \\ -1 & -7/2 \end{bmatrix} \quad G = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad H = \begin{bmatrix} 2 & 3 \end{bmatrix} \quad D = [0]$$

⇒ COMPUTE THE SYSTEM T.F.

$$W(z) = H(zI - F)^{-1} \cdot G$$

$$\cdot (zI - F) = \begin{bmatrix} z-1 & -9/2 \\ 1 & z+7/2 \end{bmatrix}$$

$$\cdot (zI - F)^{-1} = \frac{1}{\det(\cdot)} \begin{bmatrix} z+7/2 & 9/2 \\ -1 & z-1 \end{bmatrix} = \frac{1}{z^2 + 5/2z + 1} \begin{bmatrix} z+7/2 & 9/2 \\ -1 & z-1 \end{bmatrix}$$

SO WE HAVE:

$$W(z) = \frac{1}{z^2 + 5/2z + 1} \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} z+7/2 & 9/2 \\ -1 & z-1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{z^2 + 5/2z + 1} \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 1-z \\ z \end{bmatrix} = \boxed{\frac{z+2}{z^2 + 5/2z + 1} = W(z)}$$

⇒ DISCUSS THE STABILITY OF THE SYSTEM AND ALSO ITS OBSERVABILITY AND REACHABILITY

SYSTEM POLES: $z^2 + 5/2z + 1 = 0$

$$z_{1,2} = -\frac{5}{4} \pm \sqrt{\frac{25}{16} - 1}$$
$$= -\frac{5}{4} \pm \sqrt{\frac{9}{16}} = \begin{cases} z_1 = -1/2 \\ z_2 = -2 \end{cases}$$

THIS SYSTEM IS NOT ASYMPTOTICALLY STABLE, SINCE ONE OF THE POLES IS OUTSIDE THE UNIT CIRCLE.

HOWEVER, IF WE LOOK I/O RELATIONSHIP CLOSER:

$$W(z) = \frac{(z+2)}{(z+1/2)(z+2)} = \frac{1}{z+1/2} \Rightarrow \text{THE SYSTEM IS EXTERNALLY ASYMPTOTICALLY STABLE}$$

↓

SINCE THERE IS AN HIDDEN PART $\Rightarrow$ THE SYSTEM IS NOT COMPLETELY OBSERVABLE / REACHABLE

LET'S COMPUTE THE OBSERVABILITY AND REACHABILITY MATRICES:

$$O = \begin{bmatrix} H \\ HF \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ [2 \ 3] \begin{bmatrix} 1 & 9/2 \\ -1 & -7/2 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & -3/2 \end{bmatrix}$$

$$\det O = -3 + 3 = 0 \Rightarrow \text{THE SYSTEM IS NOT COMPLETELY OBSERVABLE}$$

$$R = \begin{bmatrix} G & F \cdot G \end{bmatrix} = \begin{bmatrix} -1 & \begin{bmatrix} 1 & 9/2 \\ -1 & -7/2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} -1 & 7/2 \\ 1 & -5/2 \end{bmatrix}$$

$$\det R = 5/2 - 7/2 = -1 \neq 0 \Rightarrow \text{THE SYSTEM IS COMPLETELY REACHABLE}$$

$\Rightarrow$ COMPUTE THE FIRST 4 SAMPLES OF THE SYSTEM IMPULSE RESPONSE
 $\hookrightarrow k=0,1,2,3$

$$① \quad W(z) = \frac{1}{z+1/2} = \frac{z^{-1}}{1+1/2 z^{-1}}$$

$$\begin{array}{l|l} \begin{array}{l} z^{-1} \\ z^{-1} + 1/2 z^{-2} \\ \hline -1/2 z^{-2} \\ -1/2 z^{-2} - 1/4 z^{-3} \\ \hline 1/4 z^{-3} \\ \vdots \end{array} & \begin{array}{l} 1 + 1/2 z^{-1} \\ \hline z^{-1} - 1/2 z^{-2} + 1/4 z^{-3} \end{array} \end{array}$$

$$w(0) = 0$$

$$w(1) = 1$$

$$w(2) = -1/2$$

$$w(3) = 1/4$$

2. COMPUTE THE RECURSIVE EQUATION ASSOCIATED TO $w(t)$ AND APPLY IT WHEN

$$u(t) = \begin{cases} t=0 & u=1 \\ t>0 & u=0 \end{cases}$$

⇒ COMPUTE ONE POSSIBLE STATE SPACE REPRESENTATION, STARTING FROM THE I.R. DATA

STEP # 1

$$H_1 = [w(1)] = [1] \rightarrow \det H_1 \neq 0$$

$$H_2 = \begin{bmatrix} w(1) & w(2) \\ w(2) & w(3) \end{bmatrix} = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1/4 \end{bmatrix} \rightarrow \det H_2 = \frac{1}{4} - \frac{1}{4} = 0$$

⇓
 H_2 IS NOT FULL RANK!

⇓
SYSTEM ORDER = 1

STEP # 2

$$H_2 = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1/4 \end{bmatrix} = \begin{bmatrix} \boxed{1} \\ \boxed{-1/2} \end{bmatrix} \begin{bmatrix} 1 & -1/2 \end{bmatrix}_{1 \times 2}$$

2×1

STEP # 3

$$H = [1]$$

$$G = [1]$$

$$\boxed{} = \boxed{} \cdot F \Rightarrow F = \boxed{} \boxed{}^{-1}$$

$$F = [-1/2]$$

ADDITIONAL CHECK:

$$w(z) = H(zI - F)^{-1} \cdot G = \frac{1}{z + 1/2}$$

SOLUTION OF EXERCISE 2

THE SAMPLES OF THE IMPULSE RESPONSES CAN BE COMPUTED USING THE RECURSIVE EQUATION ASSOCIATED TO $W(z)$:

$$y(t) = W(z) u(t) = \frac{z^{-1}}{1 + \frac{1}{2} z^{-1}} u(t)$$

$$u(t) = \begin{cases} 1 & t=0 \\ 0 & t \geq 1 \end{cases}$$

$\Downarrow$

$$y(t) [1 + \frac{1}{2} z^{-1}] = z^{-1} u(t)$$

$\Downarrow$

$$y(t) = -\frac{1}{2} y(t-1) + u(t-1)$$

RECURSIVE EQUATION

LET'S USE THE RECURSIVE EQUATION TO COMPUTE THE IMPULSE RESPONSE:

$$w(0) = y(0) = -\frac{1}{2} y(-1) + u(-1) = 0$$

$$w(1) = y(1) = -\frac{1}{2} y(0) + u(0) = 1$$

$$w(2) = y(2) = -\frac{1}{2} y(1) + u(1) = -\frac{1}{2}$$

$$w(3) = y(3) = -\frac{1}{2} y(2) + u(2) = \frac{1}{4}$$

SAME VALUES AS BEFORE!

EXERCISE MA(m)

CONSIDER A GENERIC MA(m) PROCESS:

$$y(t) = c_0 \cdot e(t) + c_1 \cdot e(t-1) + \dots + c_m \cdot e(t-m) \quad e(t) \sim \mathcal{WN}(0, 1)$$

⇒ IS THE PROCESS STATIONARY?

$y(t)$ IS STATIONARY IFF

• $e(t)$ IS STATIONARY ✓

• $W(z)$ IS ASYMPTOTICALLY STABLE ✓

BACKWARD SHIFT
PROPERTY

$$y(t) = W(z) \cdot e(t) = \underbrace{[c_0 + c_1 z^{-1} + \dots + c_m z^{-m}]}_{W(z) \text{ FIR, ALL POLES IN } z=0} \cdot e(t)$$

$W(z)$ FIR, ALL POLES IN $z=0$

↓

ALL THE POLES OF $W(z)$ ARE INSIDE THE UNIT CIRCLE

↓

$W(z)$ IS A.S.

THEN $y(t)$ IS STATIONARY. THIS HOLDS FOR ANY MA PROCESS!

⇒ PROVIDE A STOCHASTIC DESCRIPTION OF THE PROCESS $y(t)$

• MEAN VALUE $y(t)$

$$\begin{aligned} E[y(t)] &= E[c_0 e(t) + c_1 e(t-1) + \dots + c_m e(t-m)] \\ &= c_0 E[e(t)] + c_1 E[e(t-1)] + \dots + c_m E[e(t-m)] \\ &= c_0 \cdot 0 + c_1 \cdot 0 + \dots + c_m \cdot 0 = 0 \\ &= \sum_{i=0}^m [c_i] \cdot \mu_e \end{aligned}$$

DE-BIASED PROCESS

• COVARIANCE FUNCTION $\gamma(z) = E[(y(t) - \mu_y)(y(t-z) - \mu_y)]$

✓ VARIANCES OF $y(t)$

$$\gamma(0) = E[y(t)^2] = E[(c_0 e(t) + c_1 e(t-1) + \dots + c_m e(t-m))^2]$$

$$= c_0^2 E[e^2(t)] + c_1^2 E[e^2(t-1)] + \dots + c_m^2 E[e^2(t-m)] +$$

CROSS PROD. → $+ 2c_0 \cdot c_1 E[e(t) \cdot e(t-1)] + \dots + 2c_0 c_m E[e(t) \cdot e(t-m)]$

$$= [c_0^2 + c_1^2 + \dots + c_m^2] \cdot \lambda_e^2 + 0 + 0 + \dots + 0.$$

$$\gamma(0) = \left[\sum_0^m c_i^2 \right] \cdot \lambda_e^2$$

$$\gamma(1) = \mathbb{E}[y(t) \cdot y(t-1)]$$

$$= \mathbb{E}[(c_0 e(t) + c_1 e(t-1) + \dots + c_m e(t-m)) (c_0 e(t-1) + c_1 e(t-2) + \dots + c_m e(t-m-1))]$$

$$= c_0 \cdot c_1 \cdot \mathbb{E}[e^2(t-1)] + \dots + c_{m-1} \cdot c_m \cdot \mathbb{E}[e^2(t-m)] + 0$$

$$= \left[\sum_1^m c_{i-1} \cdot c_i \right] \cdot \lambda_e^2$$

$$\gamma(2) = \mathbb{E}[y(t) \cdot y(t-2)]$$

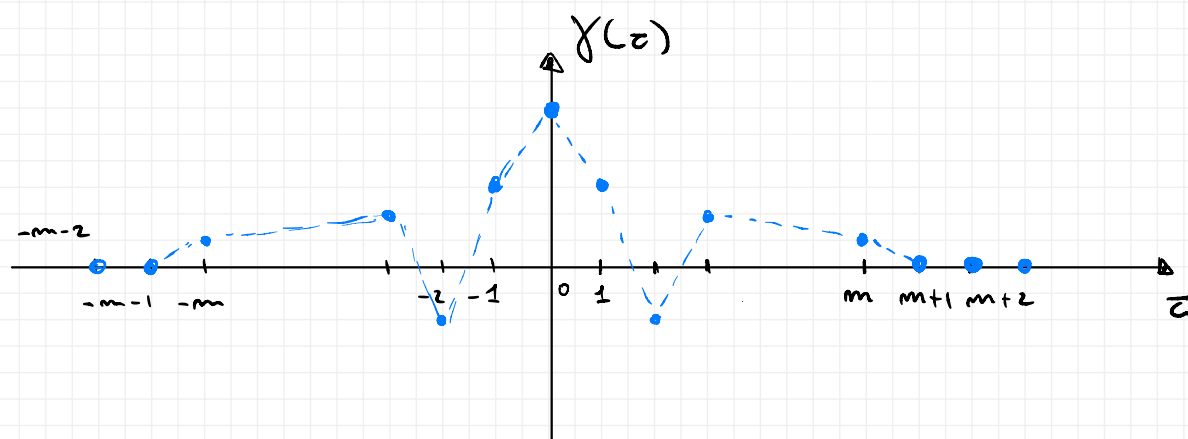
$$= \mathbb{E}[(c_0 e(t) + c_1 e(t-1) + \dots + c_m e(t-m)) (c_0 e(t-2) + c_1 e(t-3) + \dots + c_m e(t-m-2))]$$

$$= \left[\sum_2^m c_{i-2} \cdot c_i \right] \lambda_e^2$$

$$\gamma(m) = c_0 \cdot c_m \lambda_e^2$$

$$\gamma(z > m) = 0$$

MA PROCESSES HAVE A "FINITE" COVARIANCE FUNCTION



Exercise - AR(1)

Consider a generic AR(1) process:

$$y(t) = a \cdot y(t-1) + e(t)$$

$$e(t) \sim \text{WN}(0, \lambda_e^2)$$

→ Is this process $y(t)$ stationary?

- $e(t)$ is a SSP ✓
- $W(z)$ is A.S.

$$W(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

POLES OF $W(z)$ are: $z = a \Rightarrow$ This process $y(t)$ is STATIONARY only if $|a| < 1$

→ PROVIDES A STOCHASTIC DESCRIPTION OF $y(t)$

• MEAN VALUES

$$\begin{aligned} \textcircled{1} \quad \mathbb{E}[y(t)] &= \mathbb{E}[a \cdot y(t-1) + e(t)] \\ &= a \mathbb{E}[y(t-1)] + \mathbb{E}[e(t)] \end{aligned}$$

APPLY THE DEFINITION

$$\mu_y = a \cdot \mu_y + \mu_e$$

$$\mu_y = \frac{1}{1-a} \cdot \mu_e = 0$$

DC GAIN

$$\textcircled{2} \quad \mu_y = W(z=1) \cdot \mu_e$$

→ THIS FORMULA APPLIES ALWAYS

$$\mu_y = \frac{1}{1-a} \cdot \mu_e$$

• COVARIANCE FUNCTION $\gamma(\tau)$

$$\gamma(0) = \mathbb{E}[y^2(t)] = \mathbb{E}[(a \cdot y(t-1) + e(t))^2]$$

$$= a^2 \cdot \mathbb{E}[y^2(t-1)] + \mathbb{E}[e^2(t)] + 2 \cdot a \cdot \mathbb{E}[y(t-1) \cdot e(t)]^*$$

$$\gamma(0) = a^2 \cdot \gamma(0) + \lambda_e^2 + 0$$

$$\begin{aligned}
 * \quad \mathbb{E}[y(t-1) \cdot e(t+1)] &= \mathbb{E}[(a \cdot y(t-2) + e(t-1)) \cdot e(t+1)] = \\
 &\quad \left(\begin{array}{l} y(t-1) = f(e(t-1), e(t-2), \dots, e(t-p)) \end{array} \right) \\
 &= a \mathbb{E}[y(t-2) \cdot e(t+1)] + \mathbb{E}[e(t-1) \cdot e(t+1)]
 \end{aligned}$$

$$\gamma(0) = \frac{1}{1 - a^2} \cdot \lambda_e^2$$

$$\begin{aligned}
 \gamma(1) &= \mathbb{E}[y(t+1) \cdot y(t-1)] = \mathbb{E}[(a \cdot y(t-1) + e(t+1)) \cdot y(t-1)] \\
 &= a \mathbb{E}[y^2(t-1)] + \mathbb{E}[e(t+1) \cdot y(t-1)]
 \end{aligned}$$

$$\gamma(1) = a \cdot \gamma(0) + 0$$

$$\gamma(1) = a \cdot \gamma(0)$$

$$\gamma(2) = \mathbb{E}[y(t+1) \cdot y(t-2)] = \mathbb{E}[(a \cdot y(t-1) + e(t+1)) \cdot y(t-2)]$$

$$\gamma(2) = a \cdot \mathbb{E}[y(t-1) \cdot y(t-2)] + \mathbb{E}[e(t+1) \cdot y(t-2)]$$

$$\gamma(2) = a \cdot \gamma(1) + 0$$

↓ GENERAL FORMULA

$$\gamma(z) = a \cdot \gamma(z-1), \quad z \geq 1$$

$$\gamma(0) = \frac{1}{1 - a^2} \cdot \lambda_e^2$$

YULG-WALKER

EQUATIONS FOR AR(1)

